Hemjyoti Das

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Education

Ph.D in Electrical Engineering **Aug. 2022 – Present**
Vienna University of Technology (TU Wien) *Vienna, Austria*

Thesis: *Control of a Suspended Underactuated Aerial Manipulation System.*
Supervisor: Prof. Christian Ott

M.Sc in Aerospace Engineering **Sep. 2018 – July 2020**
Delft University of Technology (TU Delft) *Delft, The Netherlands*

Thesis: *Incremental Nonlinear Dynamic Inversion Control of Pneumatic Actuators.*
Supervisors: Prof. Erik-Jan van Kampen and Prof. Daan Pool

B.Tech in Electrical Engineering **Sep. 2012 – May 2016**
National Institute of Technology (NIT Silchar) *Silchar, India*

Thesis: *Visual Servoing and Hand Gesture-based Control of Mobile Robot.*

Research Experience

Ph.D. Researcher, Robotics Systems Lab (RSL), TU Wien **Aug 2022 – Present**
Supervisor: Prof. Christian Ott; Collaborator: Prof. Dongheui Lee

- **Underactuated suspended aerial platform:** Developed uSAM, a novel underactuated suspended aerial manipulator integrating a Kinova Gen3 arm with a multirotor platform that generates a planar 3-DoF wrench. Compared with fully actuated designs, it requires fewer actuators and is more energy-efficient. The platform is equipped with an onboard IMU, computing units and a 2-DoF camera tracking system.
- **Hierarchical task-space control:** Extended prioritized multi-task control to the underactuated uSAM, where control action cannot be exerted independently on all degrees of freedom. Formulated the controller as an optimization problem and validated it for trajectory tracking with the end-effector and aerial platform.
- **Passivity during physical interaction:** Augmented the hierarchical controller with an energy tank to restore passivity when the uSAM is subject to external forces. Analysed the resulting trade-off with control performance arising from the underactuation and the task hierarchy, and validated it on pipe-inspection tasks.
- **Residual reinforcement learning:** Designed a residual RL policy on top of a model-based output dynamics controller that computes both the planar platform wrench and the manipulator joint torques. The learned residuals make the nominal controller robust to unmodelled dynamics and sensor noise, while incorporating the uncontrolled nullspace dynamics into the policy to dampen its influence on the system output.
- **Swing-up, grasping and throwing maneuvers:** Augmented a hierarchical model-based controller with an RL agent to perform swing-up maneuvers of the suspended aerial manipulator, allowing it to perch at locations unreachable with the platform's thrust alone. Combined the swing-up with aggressive maneuvers such as grasping and throwing of objects, targeting crane-based applications on construction sites.
- **Crane-load stabilization:** Developed control approaches for a cable-suspended multirotor platform carrying a slung load for safe heavy-load transport with cranes. Designed a composite whole-body controller based on singular perturbation theory exploiting the system's three time-scale dynamics, together with a shared-control and an operational-space variant. Carried out stability and robustness analysis against wind, sensor noise and model uncertainty, relying only on onboard sensors, and validated in two different lab setups.

- **Hardware-in-the-loop simulation:** Developed an impedance-matching based HILS approach in which a fixed-base manipulator reproduces the end-effector dynamics of a target vehicle-manipulator system (VMS), including its redundant nullspace dynamics, in a real-time simulation framework. Contact wrenches come from real physical interaction rather than simulation, which suits systems such as orbital robots that cannot be tested after launch. Validated on contact tasks with a suspended aerial manipulator as the target.

Research Engineer, University of Twente **Aug 2020 – July 2022**
Supervisor: Prof. Antonio Franchi

- **Human intention prediction using moving horizon estimation:** Formulated a moving horizon estimation (MHE) problem to identify the parameters of a human-pilot model and predict the pilot's commands for a quadrotor, allowing off-nominal pilot behaviour to be detected and providing the predictions needed by a mixed-initiative MPC (MI-MPC) that blends human and autonomous inputs for shared autonomy. Also developed a bi-level optimization framework that tunes the weights of a QP-based controller so that its commands mimic those of the human, demonstrated on a double integrator and extended to UAV dynamics.
- **Cooperative active perception with heterogeneous multi-UAV systems:** Contributed to an NMPC-based cooperative control framework for active information acquisition, in which multiple multirotor UAVs minimize the uncertainty of a cooperative Kalman filter estimating the pose of a moving feature. The controller uses a full nonlinear model down to the motor level, including torque limits, and supports heterogeneous quadrotors and tilted-propeller multirotors with generic range-and-bearing sensors. Validated in Gazebo simulations and real experiments, and supervised a master's thesis student as part of this project.
- **Aerial human-robot handover and low-level motor control:** Developed the Gazebo simulation environment for an ergonomics-aware NMPC that lets a multirotor aerial robot autonomously hand over an object to a human coworker. The controller accounts for human ergonomics, motion prediction and collision safety. Also implemented D-Shot-based closed-loop speed control of brushless DC motors for multirotors.

Master's Thesis, TU Delft **Sep 2019 – July 2020**
Supervisors: Prof. Erik-Jan van Kampen and Prof. Daan Pool

- **Incremental nonlinear dynamic inversion control of pneumatic actuators:** Developed a cascaded control approach, consisiting of an inner-loop incremental nonlinear dynamic inversion (INDI) controller for force tracking and an outer-loop NDI controller for piston position, relying less on plant-model information and thus robust to model mismatch and external disturbances. Benchmarked the approach against a PID controller under realistic sensor noise and varying external loads.

Research Intern, German Aerospace Center (DLR) **April 2019 – August 2019**
Supervisor: Dr. Min Jun Kim

- **Closed-chain dynamic modelling of a suspended aerial manipulator:** Developed the closed-chain dynamic model of SAM, DLR's cable-Suspended Aerial Manipulator carrying a KUKA arm, using a Lagrangian formulation with constraint equations. Unlike the earlier open-chain model, this formulation includes the winches as prismatic actuators, extending the actuation available to the platform. Further, implemented PD control with gravity compensation and kinematic inversion for joint tracking of the KUKA arm and winches, together with a damping controller for the platform oscillations.

Project Assistant, Indian Institute of Science (IISc), Bangalore **Jan 2017 – July 2018**
Supervisor: Prof. Radhakant Padhi

- **Bio-inspired landing of a quadrotor with improved state estimation:** Implemented a bio-inspired guidance method based on Tau theory for safe and smooth indoor landing of an open-source quadrotor, closing motion gaps with zero relative velocity and acceleration at touchdown. Developed an extended Kalman filter fusing monocular SLAM with inertial navigation (INS) data to obtain accurate and fast estimates of the inertial position and translational velocities, aided by a depth camera (Microsoft Kinect) for precise landing.

Project Assistant, Indian Institute of Technology (IIT), Bombay **Aug 2016 – Nov 2016**
Supervisor: Prof. Hemendra Arya

- **Propulsion system analysis for quadrotor flight endurance:** Analyzed the electric propulsion system of a quadrotor to maximize hover time through the selection of brushless motors, propellers and batteries, using a three-phase DC motor model and static propeller equations. Built a MATLAB-interfaced static testbed to characterize the thrust, torque and efficiency of different motor-propeller combinations.

Publications

- **H. Das** et al., "Constrained Interaction Control of a Suspended Aerial Manipulator for Inspection in Tilted Configurations," *In preparation.*
- **H. Das** et al., "Task Space Control of an Underactuated Suspended Aerial Manipulator (uSAM)," *In preparation.*
- **H. Das**, D. Lee and C. Ott, "Residual Learning-Based Output Dynamics Control of an Underactuated Suspended Aerial Manipulator," *Under review.*
- F. Kong, G. Zambella, **H. Das**, G. Grioli, A. Bicchi and C. Ott, "Task-Informed Reachability Map for Mobile Manipulation," *Under review.*
- **H. Das**, S. Deimel and C. Ott, "Energy-based Swing-up and Grasping Maneuver using Suspended Aerial Manipulator," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.*
- **H. Das**, G. Zambella and C. Ott, "Whole-Body Stabilization of a Cable-Suspended Multirotor Platform Carrying a Slung Load," *IEEE Transactions on Automation Science and Engineering (T-ASE), vol. 22, 2025.*
- **H. Das** and C. Ott, "Partial Feedback Linearization Control of a Cable-Suspended Multirotor Platform for Stabilization of an Attached Load," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.*
- **H. Das**, M.N. Vu and C. Ott, "Learning Swing-up Maneuvers for a Suspended Aerial Manipulation Platform in a Hierarchical Control Framework," *IFAC-PapersOnLine, vol. 59, no. 18, 2025.*
- **H. Das**, M.N. Vu, T. Egle and C. Ott, "Observer-based Controller Design for Oscillation Damping of a Novel Suspended Underactuated Aerial Platform," *IEEE International Conference on Robotics and Automation (ICRA), 2024.*
- **H. Das**, B.K. Sæbø, K.Y. Pettersen and C. Ott, "Hardware-in-the-Loop Simulation of Vehicle-Manipulator Systems for Physical Interaction Tasks," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.*
- M. Jacquet, M. Kivits, **H. Das** and A. Franchi, "Motor-level N-MPC for Cooperative Active Perception with Multiple Heterogeneous UAVs," *IEEE Robotics and Automation Letters (RA-L), vol. 7, no. 2, 2022.*
- G. Corsini, M. Jacquet, **H. Das**, A. Afifi, D. Sidobre and A. Franchi, "Nonlinear Model Predictive Control for Human-Robot Handover with Application to the Aerial Case," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022.*
- **H. Das**, D. Pool and E.-J. van Kampen, "Incremental Nonlinear Dynamic Inversion Control of Long-Stroke Pneumatic Actuators," *European Control Conference (ECC), 2021.*
- **H. Das**, K. Sridhar and R. Padhi, "Bio-inspired Landing of Quadrotor using Improved State Estimation," *IFAC-PapersOnLine, vol. 51, no. 1, 2018.*
- **H. Das**, "Dynamic Inversion Control of Quadrotor with a Suspended Load," *IFAC-PapersOnLine, vol. 51, no. 1, 2018.*
- **H. Das**, V. Baskaran and H. Arya, "Static Performance Analysis of Electric Propulsion System in Quadrotors," *International Conference on Control, Instrumentation, Communication and Computational Technologies (ICCICCT), 2016.*

- **H. Das**, "A Comparative Study between a Cant Angle Hexacopter and a Conventional Hexacopter," *International Conference on Control, Instrumentation, Communication and Computational Technologies (ICCICCT), 2016.*
- **H. Das**, A.S. Mazumdar, R. Dey and L. Roy, "Experimental Implementation of Fuzzy Vision-based Tracking Control of Quad-rotor," *International Workshop Soft Computing Applications (SOFA), 2016.*

Teaching Experience

Manipulation and Locomotion, TU Wien **Spring Semesters 2023, 2024, 2025**

- Designed assignments on robotic manipulation, haptics and legged locomotion, including critically damped impedance design via double diagonalization, force-closure grasping, passivity observer/controller (PO/PC) design and floating-base modelling with zero-moment point (ZMP) computation. Led student projects based on research papers on sliding-mode momentum observers for estimating external torques and joint accelerations, and on energy-tank-based impedance control with variable stiffness.

Control of UAVs, University of Twente **Spring Semester 2022**

- Set up a Gazebo-based simulation environment for multirotor UAVs and assisted students in implementing and testing their solutions to assignments on UAV actuation mechanisms and control effectiveness.

Students Supervised

TU Wien **2023 – Present**

- **Ivo Gasic** (Master's thesis): *Parameter Identification of Floating-Base Robotic Systems.*
- **Stefan Deimel** (Bachelor's thesis): *Software Framework of the uSAM: Applications for Swing-up and Grasping Operations.*
- **Philipp Jelinek** (Master's thesis): *Hooking Maneuver and Constrained Hybrid Force-Impedance Control using Suspended Aerial Manipulators.*
- **Tarik Mustedanagic** (Bachelor's thesis): *Throwing Maneuver using a Suspended Aerial Manipulator.*
- **Klaus Trimbacher** (Bachelor's thesis): *Visual Servoing for a Suspended Aerial Platform and a Robotic Manipulator.*
- **Rene Zwiletitsch** (Automation project): *Software Framework for Onboard Control of an Aerial Platform.*
- **Sandra Foith** (Master's thesis): *Control of a Suspended Aerial Platform for Contact-based Applications.*

University of Twente **2020 - 2021**

- **M.P.W. Kivits** (Master's thesis): *Heterogeneous Cooperative Control Tracking Using Nonlinear Model Predictive Control.*
- **H.O. Abdelrahma** and **M.T. Brink** (Bachelor's theses): *Software and Hardware Integration of Electronic Speed Controllers (ESC) for an Unmanned Aerial Vehicle.*

Review Activities

- **Journals:** IEEE Transactions on Robotics (T-RO), IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Automation Science and Engineering (T-ASE), IEEE Transactions on Aerospace and Electronic Systems (T-AES)
- **Conferences:** IEEE International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

References

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